1. AIM

To determine the absorption, reflection characteristics and band gap energy of a material using UV-Visible Spectroscopy.

2. APPARATUS / TOOLS REQUIRED

1. Maya 2000 Pro Spectrometer (Ocean Optics).
2. DH-2000-BAL UV-Visible-NIR Light Source.
3. Reflection probe and sample holder.
4. Glass slide and material sample.
5. Computer with Ocean Optics SpectraSuite software.
6. Origin software for data processing and Tauc plot analysis.
7. Connecting optical fibres and cables.

3. THEORY

UV-Visible spectroscopy is a characterization technique used to study the optical properties of materials by measuring their interaction with ultraviolet and visible light. When light falls on a material, a part of it may be absorbed, transmitted or reflected. These properties provide useful information about the electronic structure and optical behaviour of the material.

Absorption:

Absorption occurs when photons transfer their energy to electrons in a material, causing electronic transitions. The absorption spectrum represents the variation of absorbance with wavelength. Absorbance is related to incident and transmitted light intensity by:

$$A = \log_{10}\left(\frac{I_0}{I}\right)$$

where I_0 is the incident light intensity and I is the transmitted light intensity.

Reflection:

Reflection is the process in which light is returned from the surface of a material. The reflection spectrum represents the variation of reflected light intensity with wavelength. It provides information about the surface and optical properties of the sample.

Band Gap Energy:

The band gap is the minimum energy required to excite an electron from the valence band to the conduction band. It is an important parameter for understanding semiconductor and optical materials.

The photon energy is given by:

$$E = \frac{hc}{\lambda}$$

The band gap can be estimated using the Tauc relation:

$$(\alpha h\nu)^n = B(h\nu - E_g)$$

where α is the absorption coefficient, $h\nu$ is photon energy, B is a constant, E_g is the band gap energy and n depends on the type of electronic transition ($n = 2$ for direct allowed transition, $n = 1/2$ for indirect allowed transition). Extrapolating the linear portion of the $(\alpha h\nu)^n$ versus $h\nu$ plot to the energy axis yields the optical band gap energy E_g.

4. PROCEDURE

A. Reference Measurement (Glass Slide):

1. Open the Ocean Optics SpectraSuite software.

2. Set the spectrometer integration time (106 ms) and averaging settings.

3. Place a clean glass slide in the sample holder.

4. Take the reference measurement under dark condition by blocking the light source.

5. Turn on the light source and record the reference measurement under light condition.

B. Sample Measurement:

1. Replace the glass slide with the coated material sample.

2. Select the required measurement modes in SpectraSuite software (Scope, Absorption, Transmission, Reflection and Irradiance).

3. Record the spectra across the selected wavelength range (200-1100 nm).

4. Observe and save the wavelength versus intensity graphs.

5. Repeat the measurements if required to obtain reliable spectra.

C. Band Gap Calculation:

1. Import the recorded spectral data into Origin software.

2. Plot absorbance versus wavelength.

3. Convert wavelength into photon energy ($E = hc/\lambda$).

4. Calculate or obtain the absorption coefficient (α) as required.

5. Construct the appropriate Tauc plot ($(\alpha h\nu)^n$ versus $h\nu$).

6. Extrapolate the linear portion of the graph to the energy axis to determine the band gap energy (E_g).

D. Reflection Measurement:

1. Use the reflection probe to direct light onto the sample and collect reflected light.

2. Record the reflectance spectrum using the spectrometer.

3. Observe the variation of reflectance with wavelength.

6. OBSERVATION & CONCLUSION

Observations:

1. The UV-Visible spectroscopy setup was successfully operated using SpectraSuite software.

2. Absorption and reflection spectra were recorded over the selected wavelength range (200-1100 nm).

3. The spectral graph showed significant variation in optical response with wavelength, displaying a sharp fundamental absorption edge in the near-UV to visible region (~410-430 nm).

4. At a characteristic wavelength of 610.23 nm, the measured absorbance was recorded as 1.560 OD at an integration time of 106 ms.

5. The recorded absorption data can be used to construct a Tauc plot for estimating the optical band gap energy (E_g).

6. The band gap energy is obtained from the extrapolation of the linear region of the Tauc plot to the photon energy axis.

Conclusion:

The absorption and reflection characteristics of the material sample were studied using UV-Visible spectroscopy. The spectral data were acquired using the Maya 2000 Pro spectrometer and DH-2000-BAL balanced light source. The recorded absorption data can be further analysed using a Tauc plot to estimate the optical band gap energy of the material. Thus, the optical characterization of the material was carried out successfully.